

CO 5 AT 3

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Course Code: CSA0925
Java Level 3 Exam

Question 1:

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, a 'Secure Exit' button, the lab title 'SIMATS Online Programming Lab', the subject 'JAVA PROGRAMMING', and the user's name 'LITHISH KUMAR R' with a dropdown arrow.

QUESTIONS

LEVEL 3 — HARD

Q1 ✓
Number

Q2 ✓
Control

1/2 solved

Level 3 — Hard (H) Steps — Control Submitted

Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.

Example 1:

Input: num = 14

Output: 6

Explanation:

Step 1) 14 is even; divide by 2 and obtain 7.

Step 2) 7 is odd; subtract 1 and obtain 6.

Step 3) 6 is even; divide by 2 and obtain 3.

Step 4) 3 is odd; subtract 1 and obtain 2.

Step 5) 2 is even; divide by 2 and obtain 1.

Step 6) 1 is odd; subtract 1 and obtain 0.

Testcase 1, n = 5

2, n = 10

3, n = 12

4, n = 18

5, n = 20

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String args[]) {
4         Scanner sc = new Scanner(System.in);
5         int n;
6         int steps = 0;
7         System.out.println("Enter a number");
8         n = sc.nextInt();
9         while(n > 0) {
10             if(n % 2 == 0) {
11                 n = n / 2;
12                 steps++;
13             } else {
14                 n--;
15                 steps++;
16             }
17         }
18         System.out.println(steps);
19     }
20 }
```

Output

```
Enter a number
5
6
```

Question 2:

QUESTIONS

LEVEL 3 — NAME

Q1

Numbers

Q2

Colors

2/2 solved

Level 3 — NAME (H) Number to YWD — Numbers

✓ Submit

Write a Java Program to Convert a Given Number of Days in Terms of Years, Weeks & Days.

Sample Input/Output:

Enter the number of days:756

No. of years:2

No. of weeks:3

No. of days:5

Test cases:

1. 36

2. 36

3. 0

4. -365

5. -43

Testcase:

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         int days;
5         Scanner sc = new Scanner(System.in);
6         System.out.println("Enter the number of days:");
7         days=sc.nextInt();
8         if(days<0){
9             System.out.println("Invalid Input");
10        }
11        else{
12            int years=days/365;
13            days = days%365;
14            int weeks = days/7;
15            days = days%7;
```

Output

```
Enter the number of days:
Invalid Input
```